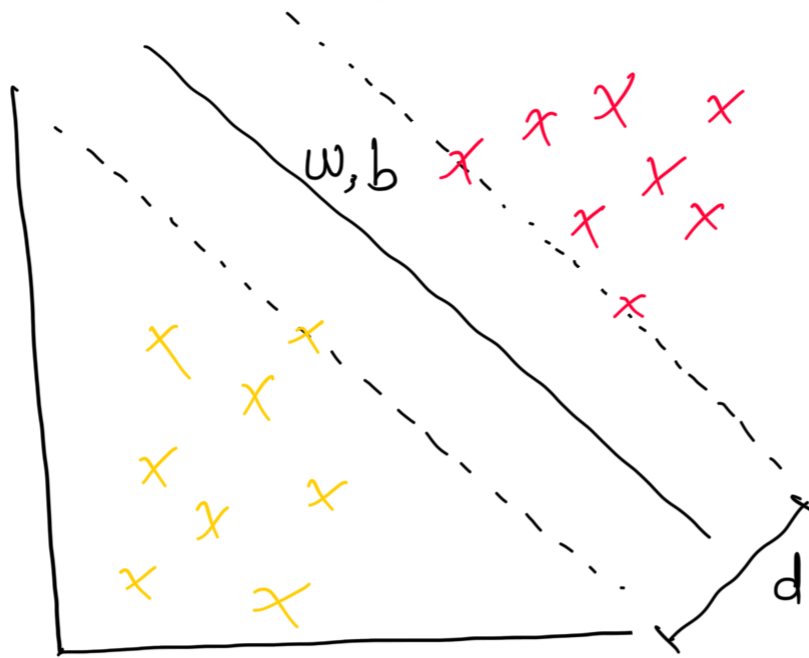
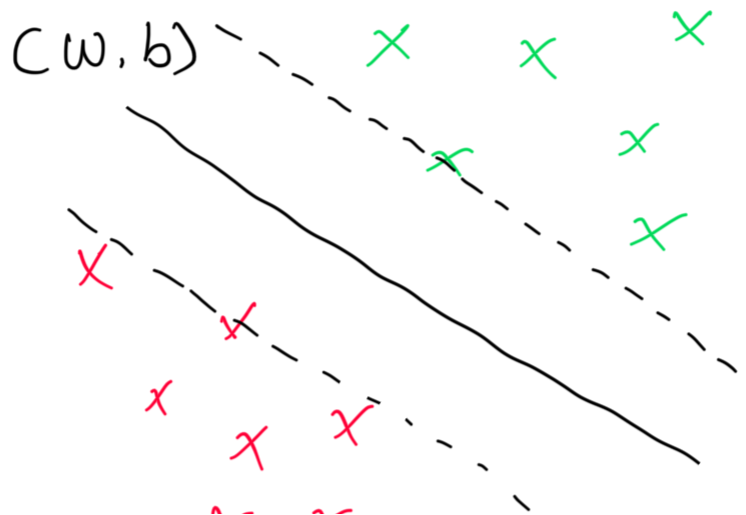


Mathematics of SVM



Decision Rule

$$\vec{w} \cdot \vec{u} + b \geq 0$$



x x

$$W^T x + b = 0$$

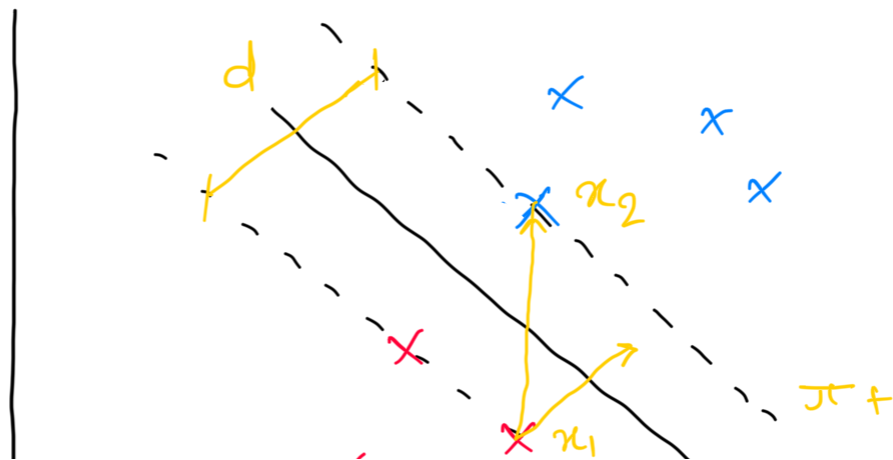
For any $\bar{x}_i$

$$\hat{y} = \begin{cases} +1 & \text{if } \vec{w} \cdot \vec{x}_i + b \geq 0 \\ -1 & \text{if } \vec{w} \cdot \vec{x}_i + b < 0 \end{cases}$$

$$y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1$$

For support vectors

$$y_i (\vec{w} \cdot \vec{x}_i + b) = 1$$





Constraint

$$y_i (\vec{w} \cdot \vec{x}_i) \geq 1$$

for support vectors

$$\boxed{y_i (\vec{w} \cdot \vec{x}_i + b) = 1}$$

$$d = \frac{(\bar{x}_2 - \bar{x}_1) \cdot \vec{w}}{\|\vec{w}\|}$$

$$= \frac{\bar{x}_2 \cdot \vec{w} - \bar{x}_1 \cdot \vec{w}}{\|\vec{w}\|}$$

$$y_i (\vec{w} \cdot \bar{x}_i + b) = 1$$

$$1 (\vec{w} \cdot \bar{x}_2 + b) = 1$$

$$\vec{w} \cdot \bar{x}_2 = 1 - b$$

$$-1 (\vec{w} \cdot \bar{x}_1 + b) = 1$$

$$= \frac{(1 - b) - (-b - 1)}{\|\vec{w}\|}$$

$$= \frac{1 - b + b + 1}{\|\vec{w}\|}$$

$$\underline{\vec{w} \cdot \vec{x}_1 = -b - 1}$$

$$= \frac{2}{\|\vec{w}\|}$$

$$\therefore \boxed{d = \frac{2}{\|\vec{w}\|}}$$

such that

$$y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1$$

Hard Margin SVM

Soft Margin SVM

$$\boxed{\frac{\|\vec{w}\|}{2} + c \sum_{i=1}^n \epsilon}$$